



Figure 1: Cloud computing layers and open source

high availability across many cloud types to provide business process continuity and disaster recovery

- Single pane and single view of all the applications across clouds are a must to ensure effective governance, security and cost management. A common process across all environments is mandated to ensure this homogeneity

Benefits of cloud adoption

Cloud adoption helps businesses to increase their flexibility to deliver IT resources, improves disaster recovery capabilities and lowers IT capital expenses. There are other benefits too.

- **Business acceleration:** The cloud helps to speed up business processes, supports collaboration, and provides cost-effective solutions to free up IT budgets for innovative, revenue-generating projects.
- **Cost reduction:** It helps in reducing operating and capital costs and improves performance, productivity and business agility via a flexible,

scalable solution. Organisations can choose the applications to move across the clouds and on-premises based on their requirements.

- **Reliability:** If one cloud service goes down, some functionality is still available to users from the other deployed clouds. Generally, one public cloud can be used as backup to another cloud.
- **Risk management:** The cloud helps in mitigating risks with a single, unified, cyber security solution.
- **Managing legacy systems:** Hybrid cloud can bridge the gap between legacy and new systems, leading to major cost savings.
- **Scalability:** Applications can scale infinitely by adopting a hybrid multi-cloud strategy while keeping the core of the business secure through on-premises hosting.

Cloud computing layers and the open source products they use

Figure 1 shows the cloud computing layers and the respective open source

products/tools/software that map to each layer of the cloud.

The layers cover the cloud market, cloud broker platform, cloud management, SaaS, PaaS, IaaS, virtualisation software/management, security, analytics, monitoring tool and hardware.

Cloud based open source technology enables enterprises to create and deploy applications with speed and consistency.

Application platforms handle complex connectivity issues behind the scenes (including automatic documentation), letting users concentrate on creating the right workflows and business logic, and also making it easier and more cost effective for organisations to manage frequent change. It is imperative to make sure that the integration solution is optimised for connecting the new applications in the cloud environments with older applications, systems, services and databases, no matter where they are located.

Open application programming interfaces (APIs): The application